

Experiment ‘p_6_1_d_process_array_first_and_last_array’

Results

June 5, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing assume : 0
Responses containing expect : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_6_1_d_process_array_first_and_last_array
Natural language statement: null
Method signature: p_6_1_d_process_array_first_and_last_array(inputs: array<int>) returns (first_and_last: array<int>)

Ensures

- `first_and_last[..] == [inputs[0], inputs[inputs.Length - 1]]`

Requires

- `inputs.Length == 10`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

p_6_1_d_process_array_first_and_last_array(inputs: array<int>) returns (first_and_last:
↪ array<int>)

The method should respect the following contract:

requires inputs.Length == 10, ensures first_and_last[..] == [inputs[0], inputs[inputs.
↪ Length - 1]]

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnNfv2ZjSA5Op50XijnM7nBO2rVX0

Model: gpt-5.5-2026-04-23

Created at: 1780661451

Finish reason: stop

Usage: 302 tokens in, and 310 tokens out

```
{  
  first_and_last := new int[2];  
  first_and_last[0] := inputs[0];  
  first_and_last[1] := inputs[inputs.Length - 1];  
}
```

Final Program

```
method p_6_1_d_process_array_first_and_last_array(inputs: array<int>) returns (  
  ↪ first_and_last: array<int>)  
  requires inputs.Length == 10  
  ensures first_and_last[..] == [inputs[0], inputs[inputs.Length - 1]]  
{  
  first_and_last := new int[2];  
  first_and_last[0] := inputs[0];  
  first_and_last[1] := inputs[inputs.Length - 1];  
}
```

Total Token Usage

Input tokens: 302

Output tokens: 310

Reasoning tokens: 246

Sum of ‘total tokens’: 612

Experiment Timings

Overall Experiment started at 1780661450554, ended at 1780661480316, lasting 29762ms (29.76 seconds)

Iteration #1 started at 1780661450618, ended at 1780661480316, lasting 29698ms (29.70 seconds)